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Perplexity

Perplexity is a commonly used metric for evaluating language models, especially autoregressive models (like GPT, LSTM, etc.). It's similar in purpose to metrics like BLEU—quantifying how well a model generates or predicts text—but its approach is probabilistic rather than based on n-gram matches.

Formal Definition:

Perplexity measures how well a model predicts a sequence of text. Mathematically, it is the exponential of the cross-entropy:

$$Perplexity = exp\left(-\frac{1}{N}\cdot\sum_{i=1}^{N} log\left[P(w_i|w_{<i})\right]\right)$$

Where:

- $w_1, w_2, ..., w_N$ are the words (or tokens) in the sequence.
- $P(w_i|w_{<i})$ is the probability assigned by the model to word w_i given its previous context.

Intuition

- Perplexity can be interpreted as the average "surprise" the model experiences when seeing the actual text.
- A perfect model would have a perplexity of 1, meaning no surprise at all.
- Higher perplexity means the model is less confident in predicting the actual tokens.

Comparison with BLEU:

Metric	Type	Measures...	Typical Range
Perplexity	Probabilistic	How probable the model finds the actual sequence	1 to ∞ (lower is better)
BLEU	N-gram-based	How similar generated text is to a reference	0 to 1 (higher is better)

Cuadro 1: Comparison between Perplexity and BLEU metrics.

- Perplexity evaluates a language model's ability to predict text.
- BLEU evaluates the quality of generated text, such as in translation or summarization, by comparing it to a reference.

When to Use Perplexity

- To evaluate autoregressive models during training.
- In tasks such as language modeling, text prediction, or pretraining LLMs.
